



Inspiring Excellence

BRAC University

Department of Computer Science and Engineering

CSE 443: Bioinformatics-I (Makeup)

Mid Exam: Summer 2025

Time: 1 hour 30 minutes

Marks: 40

Name		ID		Section	
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- What are the differences between single cell-RNA seq with dna micro-array techniques in terms of sequencing and interpretation? [2]
 - In Gibbs Sampling, after creating the profile matrix, the k-mer probability distribution does not add up to 1. Why? [2]
 - What problems might arise in Motif Finding with randomized algorithms if we have gene upstream regions with skewed distribution of nucleotides? [2]
 - Suppose you are given a motif matrix as follows:

ATCGT
GTCCA
CTAGT
CTCCT
CTCTG

For this matrix, calculate the profile and find the profile most probable k -mer of the string *ACCTGTTTAT* using that profile. [4]

- Construct a *de Bruijn* graph from the following k-mer reads: *AGC*, *ATG*, *CAG*, *CTG*, *GCA*, *GCT*, *TGA*, *TGC*. From the constructed *de Bruijn* graph try to construct the genome using the Euler Path algorithm. [6]
 - The model of de-bruijn graph used for genome assembly taught in your class had some unrealistic assumptions. What are those? How do the bio-informaticians tackle these problems in real? [4]
- For the following pattern and text, illustrate the working principle of the Boyer-Moore algorithm and show how much is saved using the Boyer-Moore algorithm compared to the naive exact matching. Use both good suffix and bad character rules. [6]
Text: TACGACGTACGTAGACGTACG
Pattern: ACGTAC
 - What are the ways to reduce the size of the index table in reference genome based read mapping or alignment? How does that affect search efficiency? [4]

4. (a) Suppose you are given the following data for gene expression analysis.

Time	Time X	Time Y	Time Z
Gene 1	10	8	10
Gene 2	10	0	9
Gene 3	4	8	3
Gene 4	7	8	3
Gene 5	10	2	3

Soft k -means algorithm uses fuzzy or soft membership and weighted update of the centroids according to the following two formula.

membership	$w_{ik} = \frac{e^{-\beta\ x_i - \mu_k\ ^2}}{\sum_{j=1}^K e^{-\beta\ x_i - \mu_j\ ^2}}$
centroid update	$\mu_k = \frac{\sum_{i=1}^N w_{ik} x_i}{\sum_{i=1}^N w_{ik}}$

Given two cluster centers, $\mu_1 = (10, 8, 10)$, $\mu_2 = (4, 8, 3)$ and the data points, perform the following considering $\beta = 0.5$:

- i. Find the membership of each of these data points to the clusters. [3]
 - ii. Update the cluster centers. [3]
- (b) How does the asymmetric nature of KL-divergence contribute to the effectiveness of stochastic neighbor embedding? [4]